2.05 Statistical Hypothesis Testing				
2.05a	The language of hypothesis testing	a) Understand and be able to use the language of statistical hypothesis testing, developed through a binomial model: null hypothesis, alternative hypothesis, significance level, test statistic, 1-tail test, 2-tail test, critical value, critical region, acceptance region, *p*-value. *Hypotheses should be stated in terms of parameter values (where relevant) and the meanings of symbols should be stated. For example,* *"$\mathrm{H}_0 : p = 0.7$, $\mathrm{H}_1 : p \neq 0.7$, where p is the population proportion in favour of the resolution".* *Conclusions should be stated in such a way as to reflect the fact that they are not certain. For example,* *"There is evidence at the 5% level to reject H_0. It is likely that the mean mass is less than 500 g."* *"There is no evidence at the 2% level to reject H_0. There is no reason to suppose that the mean journey time has changed."* *Some examples of incorrect conclusion are as follows:* *"H_0 is rejected. Waiting times have increased."* *"Accept H_0. Plants in this area have the same height as plants in other areas."*		MO1

OCR Ref.	Subject Content	Stage 1 learners should ...	Stage 2 learners additionally should ...	DfE Ref.
2.05b 2.05c	Hypothesis test for the proportion in a binomial distribution	b) Be able to conduct a statistical hypothesis test for the proportion in the binomial distribution and interpret the results in context. c) Understand that a sample is being used to make an inference about the population and appreciate that the significance level is the probability of incorrectly rejecting the null hypothesis. *Learners should be able to use a calculator to find critical values.* *Includes understanding that, where the significance level of a test is specified, the probability of the test statistic being in the rejection region will always be less than or equal to this level.* [*The use of normal approximation is excluded*.]		MO2

OCR Ref.	Subject Content	Stage 1 learners should ...	Stage 2 learners additionally should ...	DfE Ref.
2.05d 2.05e	Hypothesis test for the mean of a normal distribution		d) Recognise that a sample mean, $\overline{X}$, can be regarded as a random variable. *Learners should know and be able to use the result that if* $X \sim \mathrm{N}(\mu, \sigma^2)$ *then* $\overline{X} \sim \mathrm{N}\left(\mu, \frac{\sigma^2}{n}\right)$. [*The proof is excluded.*] e) Be able to conduct a statistical hypothesis test for the mean of a normal distribution with known, given or assumed variance and interpret the results in context. *Learners should be able to use a calculator to find critical values, but standard tables of the percentage points will be provided in the assessment.* [*Test for the mean of a non-normal distribution is excluded.*] [*Estimation of population parameters from a sample is excluded*]	MO3

OCR Ref.	Subject Content	Stage 1 learners should ...	Stage 2 learners additionally should ...	DfE Ref.
2.05f 2.05g	Hypothesis test using Pearson's correlation coefficient		f) Understand Pearson's product-moment correlation coefficient as a measure of how close data points lie to a straight line. g) Use and be able to interpret Pearson's product-moment correlation coefficient in hypothesis tests, using either a given critical value or a *p*-value and a table of critical values. *When using Pearson's coefficient in an hypothesis test, the data may be assumed to come from a bivariate normal distribution.* *A table of critical values of Pearson's coefficient will be provided.* [*Calculation of correlation coefficients is excluded.*]	MO1